From Spatial to Frequency Domain: A Pure Frequency Domain FDNet Model for the Classification of Remote Sensing Images

Wei Wang, Tianle Yang, and Xin Wang

***Abstract*— In remote sensing scene classification (RSSC), a common approach to reduce processing costs is to downsample images in the spatial domain to lower resolution. However, this approach can lead to the loss of texture information. Conversely, the frequency domain analysis can effectively extract frequency variations of pixel values, aiding in distinguishing texture differences in remote sensing scenes. Therefore, it is common in RSSC to use frequency domain analysis as an auxiliary branch of neural networks to extract additional information. The essence of these approaches is to extract feature information from both the spatial and frequency domains and then fuse them together. However, this combination method does not fully leverage the benefits of frequency domain feature extraction. The approach of extracting feature information solely from the frequency domain may require further research and experimentation to explore its potential advantages and applications. This study integrates wavelet transform into downsampling operations and creatively designs a network model called the frequency domain deep learning network (FDNet), which uses frequency domain methods as the backbone of a neural network and performs feature extraction solely in the frequency domain. FDNet not only solves the problem of information loss caused by spatial downsampling but also provides a new research direction of combining frequency domain methods with neural networks, that is, using frequency domain methods as the backbone of neural networks. FDNet also provides a new research direction for RSSC, which is to extract features only in the frequency domain. FDNet has been evaluated on three public remote sensing datasets: SIRI-WHU, RSSCN7, and AID. Each dataset underwent three experiments, with average classification accuracies of 97.78%, 97.62%, and 95.92%, respectively. The results show that FDNet outperforms other advanced methods participating in the comparison of classification tasks.**

***Index Terms*— Convolutional neural network (CNN), frequency domain deep learning network (FDNet), remote sensing scene classification (RSSC), vision transformer (ViT), wavelet transform.**

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The authors are with the School of Computer and Communication Engineering, Changsha University of Science and Technology, Changsha 410114, China (e-mail: wangxin@csust.edu.cn).

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NOMENCLATURE

CNN	Convolutional neural network.
CSA	Channel self-attention.
CV	Computer vision.
CWNN	Convolutional wavelet neural network.
DWT	Discrete wavelet transform.
FCNN	Fully connected neural network.
FDNet	Frequency domain deep learning network.
HLSA	High-frequency and low-frequency self-attention.
MACW	Multihead attention with CNN and wavelet.
MDFSA	Multidimensional frequency domain self-attention.
MFT	Multimodal fusion transformer.
MSFC	Multiscale frequency domain convolution.
MSL	Multiscale local feature extraction.
RSSC	Remote sensing scene classification.
ViT	Vision transformer.
WCNNs	Wavelet CNNs.
WDTCNN	Wavelet decomposition and transformation CNN
WNN	Wavelet neural network.

I. INTRODUCTION

AS REMOTE sensing technology advances quickly, and the volume of image data available for Earth observation grows daily [1]. In RSSC, high-resolution remote sensing images provide rich and extensive data resources for research fields, such as scene classification, target recognition, and change detection [2]. Factors, such as the diversity of surface information and complex backgrounds, bring a series of challenges to the RSSC. To meet these challenges, more in-depth analysis and feature extraction are needed to effectively distinguish and identify different geographical features and targets [3]. Due to the diversity and refinement of scene categories, the similarities between remote sensing images of different categories increase, while the differences between images of the same category increase significantly. This problem makes it more difficult to correctly classify and identify remote sensing images. Therefore, the selection of more efficient photograph feature expression techniques and classification algorithms is essential to enhancing scene classification performance [4].

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When classifying remote sensing scenes in the spatial domain, downsampling operations, such as average pooling, max pooling, and convolution are usually used to reduce computational costs. However, these downsampling techniques are irreversible and may result in the loss or blurring of crucial information, which could compromise the precision and dependability of ensuing image processing and analysis operations. Conversely, for RSSC, frequency domain analysis can effectively extract frequency variations in pixel values, aiding in distinguishing textures. Furthermore, by integrating frequency domain features, RSSC models can better capture complex spatial relationships and changes in scene features, thereby improving image interpretation and classification accuracy. Incorporating frequency domain analysis into RSSC methods has been shown to enhance the performance of classification algorithms [5]. For RSSC, the typical way to combine frequency domain methods and neural networks is to use frequency domain methods as an auxiliary branch to complement features. The core idea of this method is to extract feature information from both the spatial and frequency domains of an image and then combine them. However, this combination method does not fully utilize the advantages of feature extraction in the frequency domain. The use of frequency domain methods as the backbone of neural networks and methods that extract feature information only from the frequency domain may require further research and experimentation to explore their potential advantages. To reduce the loss of detailed information in the downsampling process, while exploring the advantages and applications of frequency domain methods as the backbone of neural networks and to improve traditional spatial domain image processing methods based on the characteristics of frequency domain images, we designed a network model FDNet that only performs feature extraction and processing in the frequency domain. The following is a summary of the main contributions.

1) Addressing the issue of losing fine details in traditional downsampling and further exploring the potential advantages and applications of using frequency domain methods as the backbone of neural networks, an FDNet model that extracts features only in the frequency domain is proposed.
2) To address the issue of detail information loss during downsampling, FDNet utilizes the property of wavelet transform to resize images. Wavelet transform is applied to the downsampling process, and wave stem block and patch embedding block are designed. Compared with traditional downsampling operations, wave stem block and patch embedding block can effectively preserve texture information.
3) Due to the integration of wavelet transform into the downsampling operation, FDNet requires feature extraction in the frequency domain. Therefore, MSFC block and MDFSA block are designed based on the characteristics of frequency domain information. The MSFC block and MDFSA block perform frequency domain feature extraction at the local and multidimensional global levels, respectively.

This article is organized as follows. Section II is a related work that describes the current state of development in mainstream methods for RSSC. Section III describes the structure of FDNet, including the wave stem block, patch embedding block, MSFC block, and MDFSA block. Section IV presents the experiments, including experimental preparation, experimental results analysis, ablation experiments, and visual analysis. Section V is the conclusion. In the Nomenclature, we provide the full names of the acronyms used in this article.

II. Related Works

Currently, the mainstream methods of the RSSC network involve feature extraction in the spatial domain and feature extraction in both the spatial and frequency domains. The methods of feature extraction in the spatial domain include the networks based on the CNN structure, the networks based on the ViT structure, and the networks based on the hybrid structure of CNN and ViT. The method of feature extraction in both the spatial and frequency domains involves the combinations of the frequency domain method and neural network.

A. CNN Structure for RSSC

CNN is the mainstream architecture in CV and is widely used for various tasks and applications. The CNN models, including AlexNet [6], VGGNet [7], GoogLeNet [8], and ResNet [9], have achieved significant success. CNN usually includes convolutional layers, pooling layers, and fully connected layers [10]. The convolutional layer uses different convolutional kernels to stack convolutional operations, which helps to extract multilevel features from the input image. This method has the advantages of weight sharing and translation invariance. On the other hand, the pooling layer reduces the dimensionality of the output feature map through dimensionality reduction operations, and at the same time can reduce the training parameters of the network, helping to prevent overfitting. Therefore, CNN is well suited for handling scene classification problems with complexity and diversity [11]. In the networks based on the CNN structure for RSSC, the attention-consistent network introduced various attention techniques to comprehensively explore object information in remote sensing scene images [12]. Xiaohu et al. [11] fully combined the advantages of 2-D and 3-D CNNs in spatial domain information extraction to improve the classification accuracy of images. Huihui et al. [13] enhanced the spatial sampling position capability by adaptively adjusting the receptive fields for different targets in the scene. Because CNN structures have narrow receptive fields, methods based on them still struggle to connect long-range meanings [14].

B. ViT Structure for RSSC

Initially, the Transformer is mainly used to capture long-distance information between sequences in the field of natural language processing [15]. Subsequently, this architecture is gradually introduced into the field of CV and begins to demonstrate its ability to process image data.

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Dosovitskiy et al. [16] proposed ViT first, which used a multihead self-attention mechanism to model the contextual relationship between image pixels. In the networks based on the ViT structure for RSSC, Roy et al. [17] enhanced the ViT structure's ability to learn long-range feature correlations by integrating other multimodal data sources. Spectral query spatial introduced position encoding relative to the central pixel to highlight the relative positional relationships between the central pixel and its neighboring pixels, thereby reducing the interference of absolute spatial position information and absolute orientation information of neighboring pixels on attention calculation [18]. MFT innovatively combined the ViT with multiple modal data sources to enhance the performance of land cover classification [17]. However, the high computational cost of ViT makes it challenging to effectively balance accuracy and model complexity [14].

C. Hybrid Structure of CNN and ViT for RSSC

Compared with models that extract single features, hybrid models of CNN and ViT built with serial or parallel structures have better classification performance. In the networks based on the hybrid structure of CNN and ViT for RSSC, Wang et al. [19] extracted features of different scales and proposed a global–local feature extraction model, which had better local information and global information extraction capabilities. BFRNet used a 2-D local window self-attention block and a pyramid pool self-attention block to extract local space, channel information, and global information. Finally, global and local information were fused through the intersection of linear attention blocks [20]. Lou et al. [21] proposed a 3-D multisource domain self-attention few-shot learning model, which combines the advantages of convolution and attention mechanisms to design a 3-D feature extractor based on self-attention mechanism and deep residual convolution. These architectures operate within the spatial domain, potentially leading to the loss of distinctive high-frequency details and texture features present in remote sensing images [5].

D. Structure Combining the Frequency Domain Method and Neural Network

Image processing in the spatial domain and frequency domain are the two main approaches for image processing tasks. Since neural networks process images directly in the spatial domain, they are inherently spatial domain methods. Spatial domain and frequency domain methods have different characteristics. The combination of frequency domain methods and neural networks is one of the current research hotspots in the field of image processing [22]. WCNNs incorporated spectral analysis into CNN and also analyzed how to reformulate the convolutional and pooling layers in neural networks into generalized forms of filtering and downsampling [22]. CWNN replaced the FCNN of CNN with WNN. At the same time, the wavelet function was used as the activation function to solve the local minimum problem [23]. WDTCNN performed wavelet transform on the input image and only used low-pass filtered information for feature extraction, and the architecture also used the wavelet function to replace the rectified linear unit (ReLU) function used in the CNN architecture [24]. MACW first used a transformer to generate weights and used the weights for filtering. The filtered features were extracted through CNN and wavelet transform and then concatenated into the classifier [25]. Wave-ViT unified the wavelet transform and self-attention mechanism, and the inverse wavelet transform enhanced the output of self-attention by aggregating local context with a large receptive field [26]. The structure mentioned above, which combines frequency domain methods with neural networks, often uses frequency domain methods as an auxiliary branch of neural networks to supplement features. The image features are enhanced through preprocessing with frequency domain methods and merged with the features extracted by the neural network to improve classification accuracy. The core idea is to extract feature information from both the spatial and frequency domains and then combine them. However, this combination method does not fully utilize the advantages of feature extraction in the frequency domain. Further research and experimentation are needed to explore the potential advantages and applications of solely extracting feature information from the frequency domain.

III. Methodology

A. Overall Architecture

To address the issue of information loss during downsampling in the spatial domain, and fully utilize the advantages of feature extraction in the frequency domain, we designed a network model FDNet solely extracting frequency domain features, as shown in Fig. 1. In FDNet, the wavelet transform is integrated into the downsampling operation. Then, the frequency domain map enters the four stages of the network in sequence. The multiple scales' local characteristics are extracted from the frequency domain map in the first two stages using the MSFC block. In the third stage sequentially, the MSFC block and the MDFSA block are stacked to complete the transition from capturing local information to capturing global information. The blocks formed by sequentially stacking MSFC block and MDFSA block are referred to as MIX block, and the structure of the MIX block is shown in Fig. 1(c). In Fig. 1(c), FFN represents the feedforward neural network and LN represents the layer normalization [27]. In the last stage, only the MDFSA block is used for multidimensional global characteristic extraction. To utilize the previous information, the outputs of the second and third stages, as well as the outputs of the fourth stage, are concatenated through the Exshoutcut block and input into the classifier [28]. The Exshoutcut block expands the channels in the final fully connected layer of the classifier by using average pooling results of various scaled features from early stages, introducing inductive bias from the averaged features to enhance the final performance.

B. Wave Stem Block

Given the problem of information loss in the downsampling operation and the need to separate the image information of each frequency band for better feature extraction, we integrate

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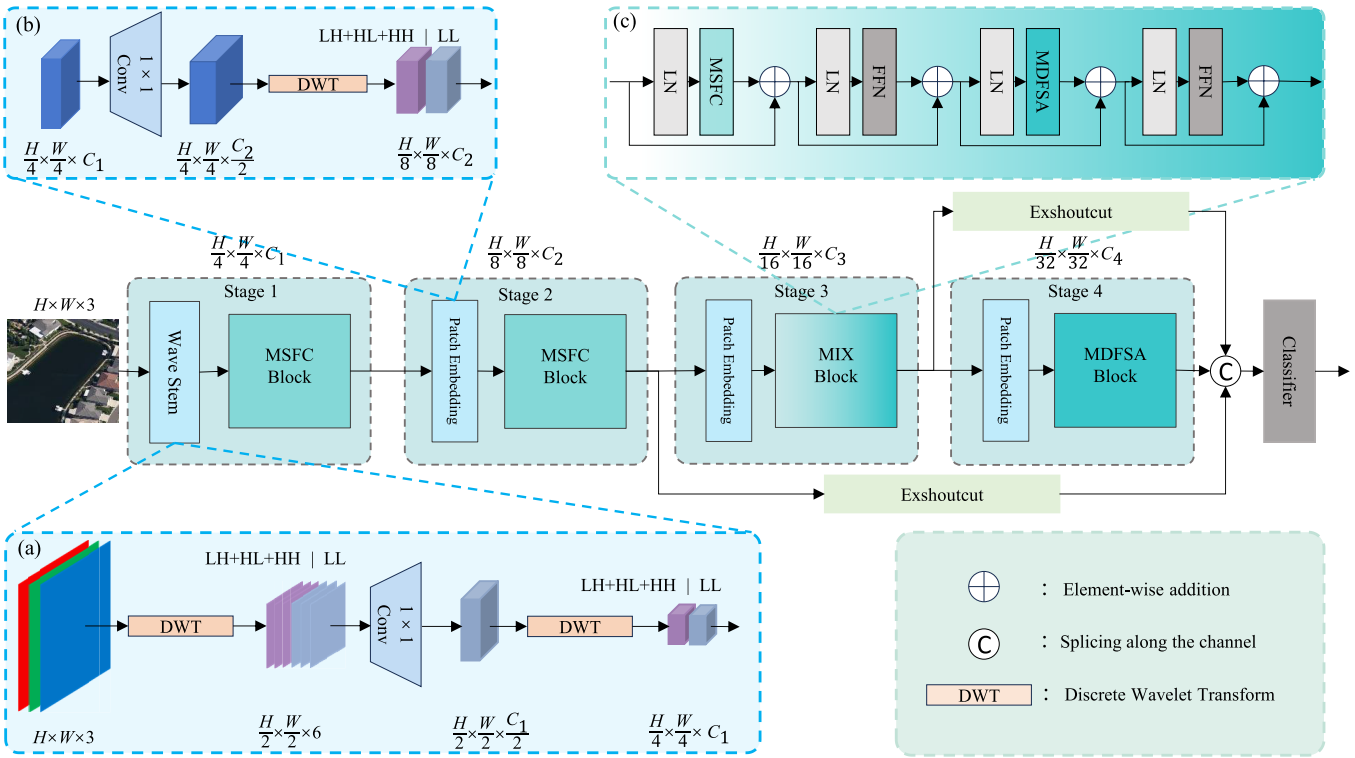


Fig. 1. Structural diagram of FDNet. (a) Structural diagram of the wave stem block. (b) Structural diagram of the patch embedding block. (c) Structural diagram of the MIX block.

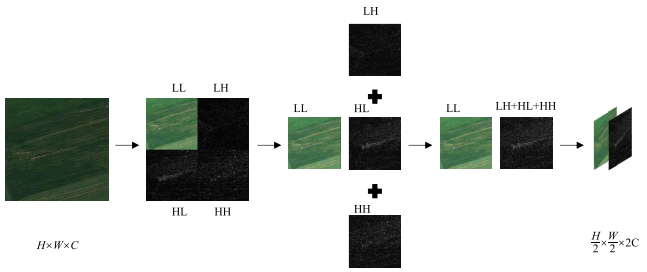


Fig. 2. Process of DWT.

the wavelet transform into the downsampling operation. The wave stem block is used to downsample the image before the MSFC block in the first stage. Its structure is shown in Fig. 1(a). X_{IN1} represents the input. After X_{IN1} undergoes DWT, a frequency domain image is obtained with half the height and width of the original image and double the number of channels. The process of DWT is shown in Fig. 2. When undergoing DWT, the image is first transformed using wavelet transform to separate the low-frequency information and high-frequency information. The three high-frequency parts of the frequency domain image are then added together, followed by concatenating the added high-frequency part and the low-frequency part along the channels. After undergoing DWT, the number of channels of the input image is expanded through a 1×1 convolutional layer. After another DWT, a frequency domain image whose height and width are both $1/4$ of the original is obtained, and the channel dimension is also expanded to the set channel value. The formula is described as follows:

$$X_{\mathrm{OUT1}} = \mathrm{DWT}(\mathrm{Conv}_{1\times1}(\mathrm{DWT}(X_{\mathrm{IN1}}))) \tag{1}$$

where DWT represents that the three high-frequency parts of the frequency domain image after wavelet transformation are first added, and then, the added high-frequency part and the low-frequency part are concatenated along the channel. $\mathrm{Conv}_{K\times K}$ represents the feature image passing through a convolutional layer of $K \times K$ size, and X_{OUT1} represents the output frequency domain diagram of the original image after passing through the wave stem block.

C. Patch Embedding Block

The patch embedding block is used to perform the next three stages of downsampling operations on the image. The difference between the patch embedding block and the wave stem block is the downsampling multiple. The wave stem block downsamples four times, while the patch embedding block downsamples two times. Its structural composition is shown in Fig. 1(b), X_{IN2} represents the input. First, the number of channels of the input image is expanded through a 1×1 convolutional layer. After DWT, a frequency domain diagram whose height and width are both $1/2$ of the original input image is obtained, and the channel dimension is also expanded to the set channel value. The formula is described as follows:

$$X_{\mathrm{OUT2}} = \mathrm{DWT}(\mathrm{Conv}_{1\times1}(X_{\mathrm{IN2}})) \tag{2}$$

where X_{OUT2} represents the output frequency domain map after passing through the patch embedding block.

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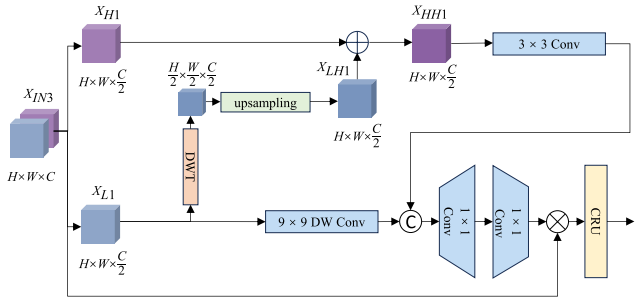


Fig. 3. MSFC block.

D. MSFC Block

In the CNN structure, the receptive field increases with the increase of the convolutional kernel, and the corresponding information extraction ability is stronger. Based on the above characteristics, the MSFC block is designed combined with the characteristics of frequency domain images. After wavelet downsampling, the MSFC block is utilized to extract multiscale local features from the frequency domain image. Its structural composition is shown in Fig. 3. First, the high-frequency and low-frequency information are separated to obtain X_{H1} and X_{L1}. X_{L1} is subjected to wavelet transformation and the high-frequency information are separated, then, upsample the high-frequency information of X_{L1} to obtain X_{LH1}. X_{HH1} is obtained by adding X_{LH1} and X_{H1}, which is the supplemented high-frequency information. In the frequency domain, low-frequency information represent the semantic information of the image, and high-frequency information represent the texture information. Therefore, an ordinary convolutional kernel of size 3×3 is used for feature extraction of the supplemented high-frequency information. Low-frequency information uses a depthwise convolutional kernel of size 9×9 for feature extraction. After feature extraction, the frequency domain images are concatenated. The channel information are fused through a 1×1 convolutional layer and then multiplied with the original frequency domain image. To further reduce redundancy along the channel dimension, the channel reconstruction block channel reconstruction unit (CRU) is used to filter channel information [29]. The channel reconstruction block CRU uses segmentation, transformation, and fusion strategies to further reduce information redundancy along the channel dimension. FDNet uses the MSFC block to implement MSL operations. The formulas are described as follows:

$$X_{\mathrm{CAT1}} = \mathrm{Concat}[\mathrm{Conv}_{3\times3}(X_{HH1}), \mathrm{DWC}_{9\times9}(X_{L1})] \quad (3)$$

$$X_{\mathrm{OUT3}} = \mathrm{CRU}(\mathrm{Conv}_{1\times1}(\mathrm{Conv}_{1\times1}(X_{\mathrm{CAT1}})) \times X_{\mathrm{IN3}}) \quad (4)$$

where X_{CAT1} represents the concatenated frequency domain image of the frequency domain image through different convolutional layers, $\mathrm{DWC}_{K\times K}$ represents the feature image passing through a depthwise convolutional layer of $K \times K$ size, X_{IN3} represents the input, X_{OUT3} represents the output frequency domain diagram after the input frequency domain diagram passes through the MSFC block, and Concat represents vectors that are concatenated along channels.

E. MDFSA Block

The MDFSA block is designed to extract multidimensional global features from the frequency domain image after wavelet downsampling. Its structural composition is shown in Fig. 4. X_{IN4} represents the input. X_{IN4} enters three branches, respectively. Branch one uses the HLSA block to extract global features in the frequency domain, branch two uses the CSA block to extract channel information in the frequency domain, and branch three uses the MSL block to extract multiscale local features. The information interaction of the three branches adopts the two-way adaptive interaction method proposed by fully adaptive transformer (FAT) [30], which first fuses global features and channel features and then fuses the fused features of the two with multiscale local features. The fusion and interaction of global features and channel features are to further integrate the information in the channel. The purpose of the fusion of the fused features and multiscale local features is to interact with local information and global information. Three-branch information interaction extracts features from three dimensions of frequency domain images. This method enables the entire network to better cope with the characteristics of frequency domain images. The formulas are described as follows:

$$X_4 = (\mathrm{sigmoid}(X_1) \times X_2) \times (X_1 \times X_2) \quad (5)$$

$$X_{\mathrm{OUT4}} = \mathrm{CRU}((\mathrm{sigmoid}(X_4) \times X_3) \times (X_4 \times X_3)) \quad (6)$$

where X_1 represents the output after the input feature mapping passes through the HLSA block, X_2 represents the output after the input feature mapping passes through the CSA block, X_3 represents the output after the input feature mapping passes through the MSL block, X_4 represents the information fusion of X_1 and X_2, and X_{OUT4} represents the information fusion output of the three branches. The following are introductions to the three branches of the MDFSA block:

1) HLSA Block: The HLSA block is different from the traditional spatial domain self-attention mechanism. The HLSA block uses different queries but uses the same value and key. the two queries are the linear mapping of the high-frequency image of the original frequency domain image and the linear mapping of the low-frequency image of the original frequency domain image. Value and key are both linear mappings of supplemented high-frequency images. The HLSA block utilizes the correlation between high-frequency and low-frequency information to achieve multilevel modeling of image features and more detailed attention weighting, thereby improving the block's ability to express image information. The implementation process first separates the high-frequency and low-frequency information to obtain X_{H2} and X_{L2}. Wavelet transform is performed on X_{L2} to separate the high-frequency information, and then, the high-frequency information of X_{L2} is upsampled to obtain X_{LH2}. X_{LH2} and X_{H2} are added to obtain X_{HH2}, which is the supplemented high-frequency information. Then, the attention weight matrices of X_{L2} and X_{H2} are got, respectively, and then, X_{L2} and X_{H2} are concatenated with the respective attention weight matrices. The structure of the HLSA block is shown in Fig. 5. The formulas are described

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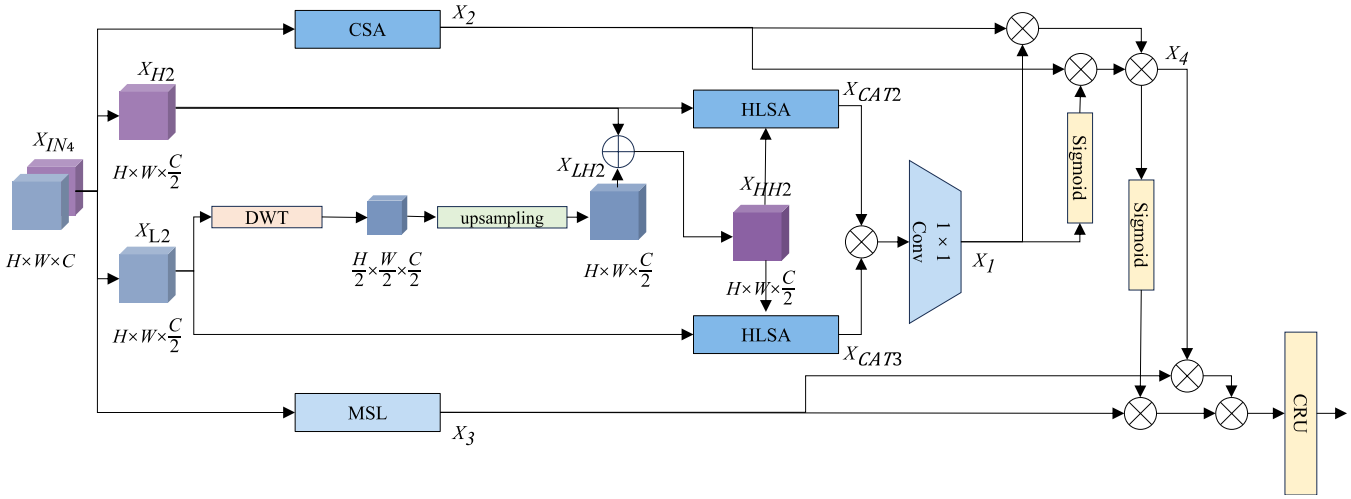


Fig. 4. MDFSA block.

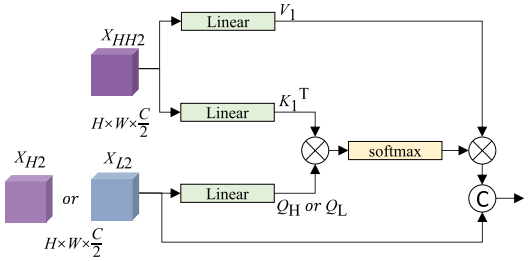


Fig. 5. HLSA block.

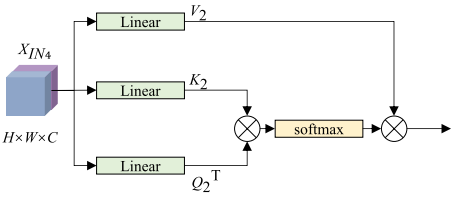


Fig. 6. CSA block.

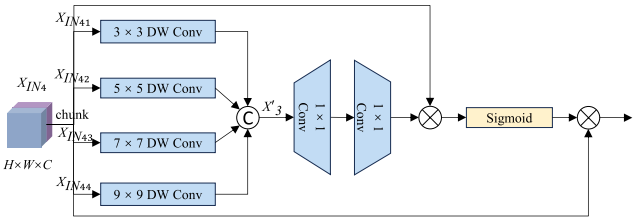


Fig. 7. MSL block.

as follows:

$$X_{\mathrm{CAT2}} = \mathrm{Concat}\left[V_1 \times \mathrm{softmax}\left(Q_H \times K_1^T\right), X_{H2}\right] \quad (7)$$

$$X_{\mathrm{CAT3}} = \mathrm{Concat}\left[V_1 \times \mathrm{softmax}\left(Q_L \times K_1^T\right), X_{L2}\right] \quad (8)$$

$$X_1 = \mathrm{Conv}_{1\times1}(X_{\mathrm{CAT2}} \times X_{\mathrm{CAT3}}) \quad (9)$$

where Q_H and Q_L represent the linear mapping of the high-frequency part of the input image and the linear mapping of the low-frequency part of the input image, and K_1 and V_1 represent the linear mappings of supplemented high-frequency images. X_{CAT2} represents the image after concatenating the attention weight matrix of X_{H2} and X_{H2}. X_{CAT3} represents the image after concatenating the attention weight matrix of X_{L2} and X_{L2}.

2) CSA Block: Since the channel dimension improvement operation will be performed multiple times during the downsampling process, the CSA module is used to extract and utilize the channel information in the frequency domain. After applying the dot product on the channel dimension between Q^T and K, a $C \times C$ size softmax scaled attention score matrix is generated to achieve the frequency domain channel information. Fig. 6 depicts the CSA block's structure, and the formula is described as follows:

$$X_2 = V_2 \times \mathrm{softmax}\left(Q_2^T \times K_2\right) \quad (10)$$

where Q_2, K_2, and V_2 represent the linear mapping of the input image.

3) MSL Block: The MSL block structure enables the network to capture local features at different scales by introducing convolutional kernels of different sizes, thereby achieving more comprehensive and multilevel information extraction. It extracts features from the dimension of local information and complements the information of the CSA block and the HLSA block to achieve multidimensional global feature extraction. Fig. 7 depicts the MSL block's construction, and the formulas are described as follows:

$$X'_3 = \mathrm{Concat}[\mathrm{DWC}_{3\times3}(X_{\mathrm{IN41}}), \mathrm{DWC}_{5\times5}(X_{\mathrm{IN42}}), \mathrm{DWC}_{7\times7}(X_{\mathrm{IN43}}), \mathrm{DWC}_{9\times9}(X_{\mathrm{IN44}})] \quad (11)$$

$$X_3 = \mathrm{sigmoid}\left(\mathrm{Conv}_{1\times1}\left(\mathrm{Conv}_{1\times1}\left(X'_3\right)\right) \times X_{\mathrm{IN4}}\right) \times X_{\mathrm{IN4}} \quad (12)$$

where X_{IN41}, X_{IN42}, X_{IN43}, and X_{IN44} are the feature maps separated by X_{IN4} along the channel, X'_3 represents the result of concatenating X_{IN41}, X_{IN42}, X_{IN43}, and X_{IN44}.

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Fig. 8. Examples of SIRI-WHU dataset.

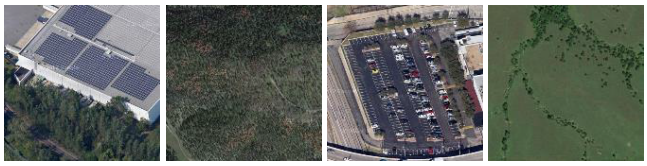


Fig. 9. Examples of RSSCN7 dataset.



Fig. 10. Examples of AID dataset.

IV. Experiments

A. Experimental Data and Setup

To verify the effectiveness of FDNet, we selected three widely used public RSSC datasets as experimental datasets. One of them is the SIRI-WHU dataset [31]. There are 12 scene classes and 2400 images in the SIRI-WHU dataset. Each class contains 200 images. The resolution of these images is 200×200, and some of the images are shown in Fig. 8. One of them is RSSCN7 dataset [32]. There are seven scene classes and 2800 images in RSSCN7 dataset. Each class contains 400 images. The resolution of these images is 400×400, and some of the images are shown in Fig. 9. The other is AID dataset [33], which has 10 000 images altogether with a resolution of 600×600 and 30 scene classes. Some of the images are shown in Fig. 10. To maintain consistency, three datasets are divided according to the ratio of 8:2; that is, a random selection of 80% of the images is utilized as the training set, while 20% of the images are used as the test set.

All of the experiments were conducted under the same hardware resources and environmental configuration. The accuracy, precision, recall, specificity, and $F1$-score are used as the evaluation criteria of the model. To enhance the robustness of the model, a series of preprocessing are performed on the experimental data. The training set images are first randomly cropped to generate images of different sizes and aspect ratios. These images are then uniformly scaled to a 224×224 resolution, randomly flipped horizontally, and finally normalized. For the test set image, we first adjust its resolution to 256×256, and then perform center cropping to the size of 224×224. To improve the training efficiency and accuracy of the model, cross-entropy loss is selected as the loss function, and the AdamW optimizer [34] is used to optimize the model parameters. Through multiple experiments, the learning rate with the best performance in each model is selected as the final learning rate. For the SIRI-WHU dataset, the initial learning rate is set to 0.001, and the weight attenuation is set to 0.05. In the comparative experiments, according to the training needs of some networks, the initial learning rate needs to be adjusted. The initial learning rate of ViT and Swin-T is set to 0.0001. The initial learning rate for multi-path vision transformer (MPViT), Wave ViT, and retentive networks meet vision transformers (RMT) is set to 0.0005. For RSSCN7 dataset, the initial learning rate is set to 0.0005, and the weight attenuation is set to 0.05. In the comparative experiments, according to the training needs of some networks, the initial learning rate needs to be adjusted. The initial learning rate of ViT and Swin-T is set to 0.0001. For AID dataset, the initial learning rate is set to 0.001, and the weight attenuation is set to 0.05. In the comparative experiments, according to the training needs of some networks, the initial learning rate needs to be adjusted. The initial learning rate of ViT, Swin-T, MPViT, and Wave-ViT is set to 0.0001. The number of iterations for each set of experiments is set to 400, and the batch size is 16.

TABLE I
Experimental Results of Modifying the Position of MIX Block on RSSCN7 Dataset

Model	MIX block location	Accuracy(%)
FDNet	**The third stage**	**97.62±0.24**
FDNet -1	All stages	96.13±0.47
FDNet -2	The fourth stage	96.73±0.23
FDNet -3	The second stage	95.95±0.30
FDNet -4	The first stage	95.12±0.59

B. Ablation Experiments

The effectiveness of each block's architecture in the FDNet network is verified by three sets of ablation experiments. All models in the ablation experiments are tested on RSSCN7 dataset and follow the same experimental settings in Section A. The first set of ablation experiments modifies the position of the MIX block of the overall architecture, and the outcomes are displayed in Table I. Experimental results show that the accuracy rate is the highest when the MIX block position is in the third stage. It can be analyzed that the transition from capturing local information to capturing global information in the third stage has the best effect and can better adapt to the characteristics of frequency domain images. The second set of ablation experiments is conducted to verify the effects of each block of the network. The outcomes are displayed in Table II. $\times$ and ✓ in Table II indicate whether the corresponding network module is enabled during the experiment. Experimental results show that FDNet achieves the highest accuracy after adding all blocks. It can be analyzed that when wave stem block and patch embedding block are discarded; that is, when the entire network does not extract features in the frequency domain, the effect is lower than the accuracy when extracting features in the frequency domain. This indicates that all block designs of FDNet are suitable for

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TABLE II
EXPERIMENTAL RESULTS OF VERIFYING THE EFFECTIVENESS OF VARIOUS NETWORK BLOCKS ON RSSCN7 DATASET

Model	Wave Stem and Patch Embedding	HLSA	High frequency supplement	CSA	Exshortcut	MSL	Accuracy(%)
FDNet	✓	✓	✓	✓	✓	✓	**97.62±0.24**
FDNet-5	×	✓	✓	✓	✓	✓	96.30±0.24
FDNet-6	✓	×	✓	✓	✓	✓	96.96±0.72
FDNet-7	✓	✓	×	✓	✓	✓	97.14±0.36
FDNet-8	✓	✓	✓	×	✓	✓	96.67±0.12
FDNet-9	✓	✓	✓	✓	×	✓	96.85±0.11
FDNet-10	✓	✓	✓	✓	✓	×	96.67±0.29

TABLE III
EXPERIMENTAL RESULTS OF WAVELET TRANSFORM BASIS FUNCTION REPLACEMENT ON RSSCN7 DATASET

Model	Wavelet transform basis function	Accuracy(%)
FDNet	**Haar Wavelet**	**97.62±0.24**
FDNet-11	Daubechies Wavelet	96.84±0.48
FDNet -12	Symlet Wavelet	96.01±1.13
FDNet -13	Coiflet Wavelet	96.43±0.71
FDNet -14	Biorthogonal Wavelet	96.19±0.60
FDNet -15	Reverse Biorthogonal Wavelet	97.14±0.18

frequency domain image processing. The third set of ablation experiments verifies the effect of wavelet transform basis function replacement, and the results are shown in Table III. The experimental results show that the entire network is optimal when Haar Wavelet is selected as the wavelet transform basis function. In signal processing, wavelets must possess properties, such as compact support, symmetry, orthogonality, and vanishing moments [35]. The symmetry of the wavelet transform basis functions directly influences the effectiveness of feature extraction in remote sensing images. Numerous images in remote sensing exhibit symmetrical and repetitive structures, such as lakes and farmlands. wavelet transform basis functions with stronger symmetry can effectively capture these local features in remote sensing images because they are more sensitive to symmetrical and repetitive structures. On the other hand, the orthogonality of wavelet transform basis functions ensures that the wavelet transform can efficiently decompose a signal into different frequency components that are independent and noninterfering. The ability to effectively separate information from different frequency bands in FDNet's structure is also crucial as the main idea behind designing FDNet is to extract features using different strategies by separating high-frequency and low-frequency information. Based on the above analysis, it is crucial for FDNet that the chosen wavelet transform basis functions possess good symmetry and orthogonality. Orthogonal wavelet transform basis functions can be used to detect edges in images but may extract a considerable amount of false edge information and are also sensitive to noise [36]. The advantages of biorthogonal wavelet transform over orthogonal wavelet transform include allowing for more degrees of freedom [37]. However, biorthogonal wavelets usually have higher computational complexity compared to orthogonal wavelets. Biorthogonal wavelets require two different filter banks, resulting in more computational operations. Based on the results of the ablation experiments and the analysis above, Haar wavelet has the advantages of lower computational complexity, good symmetry, and orthogonality. Therefore, choosing Haar wavelet as the wavelet

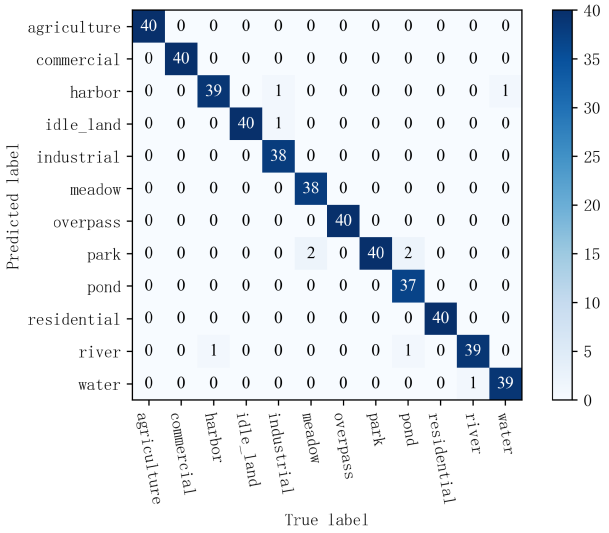


Fig. 11. Confusion matrix of FDNet on SIRI-WHU dataset.

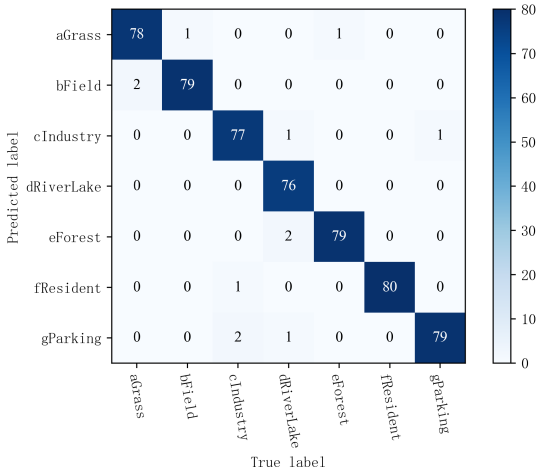


Fig. 12. Confusion matrix of FDNet on RSSCN7 dataset.

transform basis function for the FDNet model is the optimal choice. To reduce the randomness of the experimental results during the experiment, each model repeated the experiments three times.

C. Comparative Experiments

Comparative experiments are first conducted on the smaller SIRI-WHU dataset. Then, further exploration is conducted on the generalization ability of FDNet on two larger datasets.

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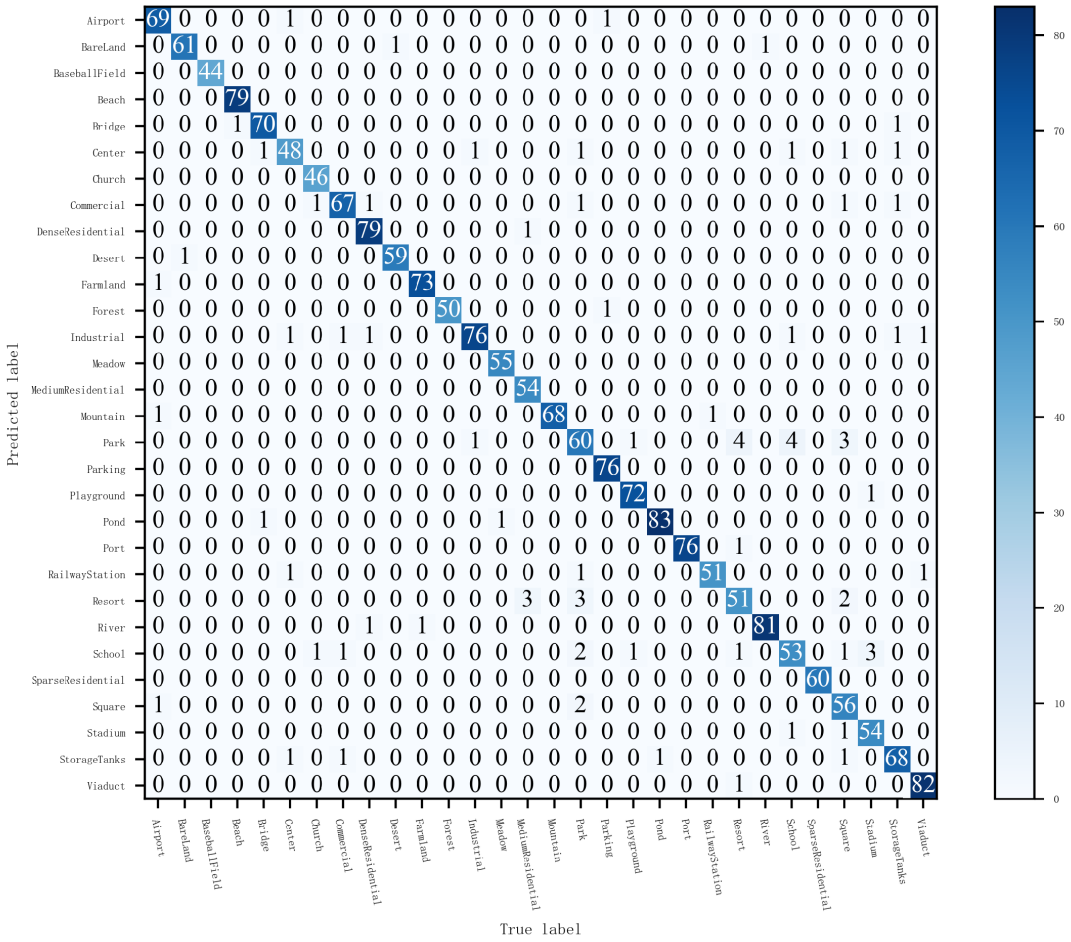


Fig. 13. Confusion matrix of FDNet on AID dataset.

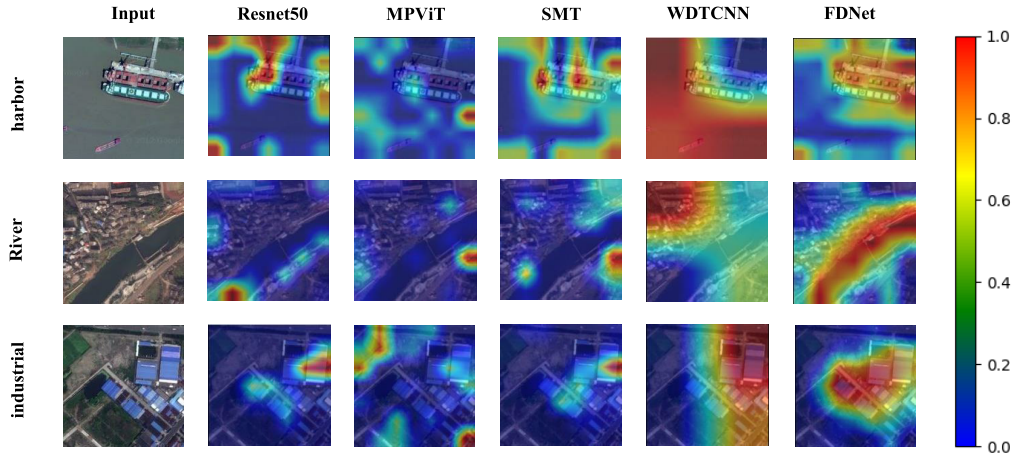


Fig. 14. Heat maps of FDNet and some models on SIRI-WHU dataset.

The compared models mainly include the networks based on the CNN structure, the networks based on the ViT structure, the networks based on the hybrid structure of CNN and ViT, and the networks based on the hybrid structure of the frequency domain method and neural network. Among them, the networks based on the CNN structure include Resnet50 [9], ConvNext [38], GCSANet [39], and DBGA-Net [40]; The networks based on the ViT structure include ViT [16], Swin-T [41], MPViT [42], and efficient multiscale transformer and cross-level attention learning (EMTCAL) [43]. The networks based on the hybrid structure of CNN and ViT include MFC2Net [44], SwiftFormer [45], SeaFormer [46], scale-aware modulation meet transformer (SMT) [47], FAT [30], and RMT [48]. The networks based on the hybrid structure of frequency domain method and neural network include WCNNs [22], Wave-ViT [26], and WDTCNN [24]. To reduce

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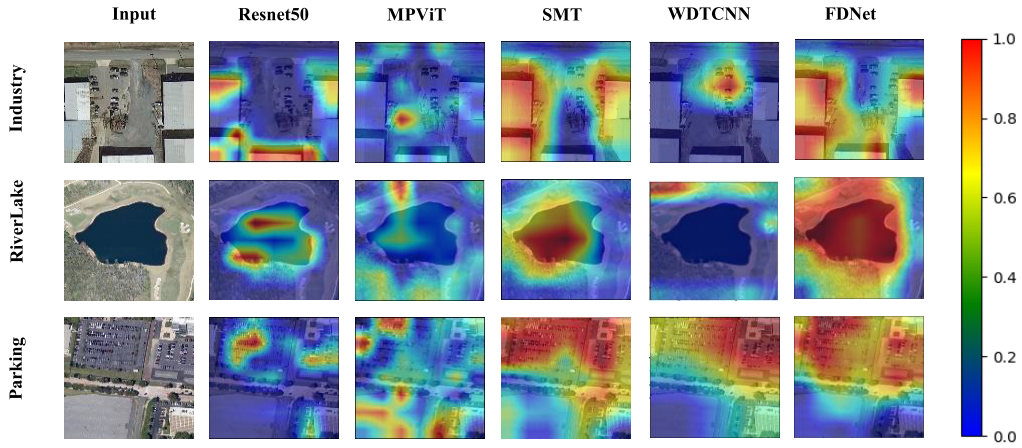


Fig. 15. Heat maps of FDNet and some models on RSSCN7 dataset.

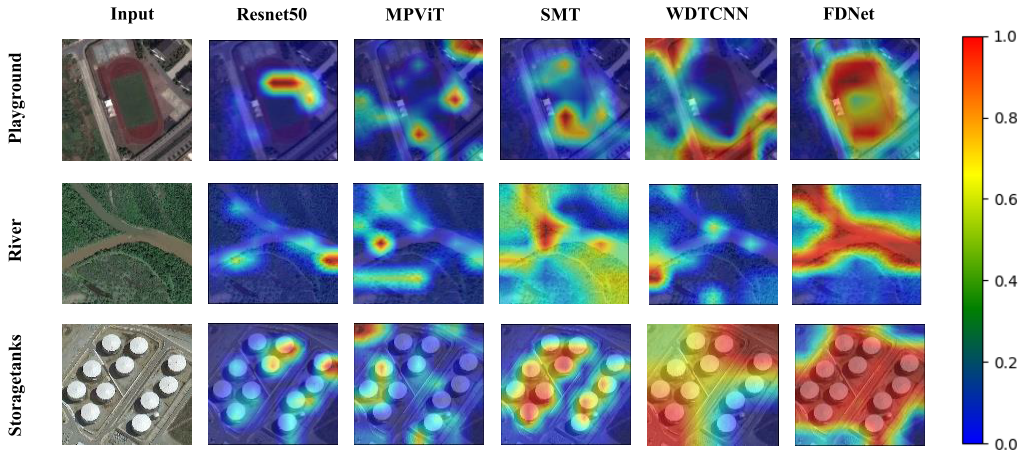


Fig. 16. Heat maps of FDNet and some models on AID dataset.

TABLE IV
PARAMETER AND COMPUTATIONAL COMPLEXITY OF ALL NETWORKS INVOLVED IN THE COMPARISON

Model	Year	Params($\times 10^6$)	FLOPs($\times 10^6$)
Resnet50 [9]	2016	25.56	4133.74
ConvNext [38]	2022	28.57	4455.53
GCSANet [39]	2022	8.11	2898.08
DBGA-Net [40]	2023	108.38	13205.20
ViT [16]	2020	85.66	16862.87
Swin-T [41]	2021	28.27	4371.89
MPViT [42]	2022	22.24	4368.58
EMTCAL [43]	2022	27.31	4233.00
MFC2Net [44]	2022	28.28	2427.60
SwiftFormer [45]	2023	11.29	1603.69
SeaFormer [46]	2023	8.66	3326.00
SMT [47]	2023	10.99	2400.50
FAT [30]	2024	7.81	1185.42
RMT [48]	2024	13.19	2321.98
WCNNs [22]	2018	5.04	791.00
Wave-ViT [26]	2022	6.09	3176.21
WDTCNN [24]	2022	31.465	1780.00
FDNet	ours	7.89	2985.46

the randomness of the experimental results during the experiment, each model repeat the experiments three times.

The parameter and computational complexity of all networks involved in the comparison are shown in Table IV. The comparisons of the classification performances of different models on SIRI-WHU dataset are shown in Table V. The comparisons of the classification performances of different models on RSSCN7 dataset are shown in Table VI. The comparisons of the classification performances of different models on AID dataset are shown in Table VII. All indicators compared are the average of three repeated experiments. As can be seen from Tables V–VII, on the SIRI-WHU, RSSCN7, and AID datasets, the classification accuracy rates of FDNet are 97.78%, 97.62%, and 95.92%, respectively. On the whole, compared with the comparative models, FDNet has excellent classification performance and is relatively stable on the three datasets. The network parameters and the computational cost are both relatively low. The network GCSANet based on the CNN structure has a classification accuracy of 94.97% on AID dataset. Although the amount of parameters and calculations are equivalent to FDNet, the classification accuracy of FDNet is 0.95% higher than that of GCSANet. The classification accuracy of the network MPViT based on the ViT structure reaches 96.31% on RSSCN7 dataset, but the classification accuracy of FDNet is 1.31% higher than that of MPViT. The classification accuracy of the network DBGA-Net based on the CNN structure reaches 96.95% on SIRI-WHU dataset, but the classification accuracy of FDNet is 0.83% higher than that of DBGA-Net. Compared with the networks based on

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TABLE V
CLASSIFICATION PERFORMANCES OF DIFFERENT MODELS ON SIRI-WHU DATASET

Model	Accuracy(%)	Precision(%)	Recall(%)	Specificity(%)	F1-score(%)
Resnet50 [9]	96.67±0.21	96.81±0.19	96.67±0.20	99.69±0.02	96.67±0.20
ConvNext [38]	93.82±0.56	93.97±0.52	93.82±0.56	99.43±0.04	93.82±0.59
GCSANet [39]	96.74±0.14	96.80±0.20	96.73±0.14	99.70±0.02	96.73±0.15
DBGA-Net [40]	96.95±0.34	97.02±0.38	96.95±0.34	99.72±0.04	96.95±0.35
ViT [16]	91.81±0.27	92.23±0.19	91.81±0.27	99.25±0.02	91.75±0.24
Swin-T [41]	95.07±0.56	95.18±0.58	95.07±0.55	99.54±0.06	95.06±0.56
MPViT [42]	96.46±0.62	96.61±0.63	96.46±0.62	99.68±0.05	96.47±0.62
EMTCAL [43]	95.48±1.19	95.67±1.12	95.49±1.18	99.59±0.11	95.49±1.18
MFC2Net [44]	96.32±0.14	96.49±0.03	96.32±0.14	99.67±0.01	96.33±0.12
SwiftFormer [45]	95.07±0.55	95.12±0.50	95.07±0.55	99.54±0.04	95.07±0.55
SeaFormer [46]	96.25±0.63	96.32±0.63	96.25±0.62	99.65±0.06	96.25±0.62
SMT [47]	93.61±0.14	93.84±0.05	93.61±0.14	99.41±0.01	93.62±0.12
FAT [30]	95.28±0.76	95.38±0.78	95.28±0.76	99.57±0.07	95.27±0.77
RMT [48]	95.69±0.14	95.80±0.17	95.69±0.14	99.61±0.01	95.71±0.14
WCNNs [22]	96.85±0.23	96.70±0.48	96.60±0.48	99.69±0.04	96.60±0.48
Wave-ViT [26]	95.07±0.35	95.23±0.32	95.07±0.35	99.55±0.04	95.08±0.36
WDTCNN [24]	96.11±0.14	96.19±0.15	96.11±0.14	99.64±0.01	96.12±0.14
FDNet	**97.78±0.14**	**97.86±0.16**	**97.78±0.14**	**99.80±0.01**	**97.78±0.14**

TABLE VI
CLASSIFICATION PERFORMANCES OF DIFFERENT MODELS ON RSSCN7 DATASET

Model	Accuracy(%)	Precision(%)	Recall(%)	Specificity(%)	F1-score(%)
Resnet50 [9]	95.24±0.47	95.28±0.45	95.24±0.52	99.23±0.08	95.23±0.46
ConvNext [38]	91.90±0.60	92.00±0.54	91.91±0.59	98.65±0.11	91.84±0.63
GCSANet [39]	95.24±0.47	95.24±0.53	95.24±0.49	99.23±0.08	95.22±0.48
DBGA-Net [40]	95.09±0.80	95.17±0.74	95.09±0.83	99.20±0.14	95.09±0.78
ViT [16]	91.61±0.35	91.72±0.35	91.59±0.35	98.61±0.05	91.59±0.35
Swin-T [41]	94.05±0.41	94.17±0.47	94.04±0.42	99.02±0.07	94.02±0.39
MPViT [42]	96.31±0.65	96.36±0.64	96.32±0.65	99.41±0.10	96.32±0.67
EMTCAL [43]	94.23±0.77	94.22±0.77	94.23±0.77	99.06±0.14	94.20±0.77
MFC2Net [44]	94.76±0.24	94.82±0.28	94.76±0.24	99.15±0.04	94.75±0.25
SwiftFormer [45]	94.58±0.24	94.69±0.38	94.59±0.22	99.13±0.03	94.55±0.27
SeaFormer [46]	94.59±0.77	94.64±0.76	94.59±0.80	99.12±0.14	94.56±0.78
SMT [47]	95.12±0.06	95.15±0.12	95.13±0.07	99.21±0.02	95.09±0.07
FAT [30]	95.77±0.30	95.82±0.28	95.77±0.29	99.32±0.05	95.77±0.30
RMT [48]	95.00±0.89	95.01±0.93	95.02±0.91	99.20±0.14	94.97±0.90
WCNNs [22]	95.12±0.59	95.16±0.64	95.12±0.59	99.21±0.09	95.11±0.60
Wave-ViT [26]	93.99±0.65	94.07±0.64	93.99±0.67	99.00±0.11	93.96±0.65
WDTCNN [24]	95.31±0.07	95.35±0.08	95.30±0.06	99.24±0.02	95.29±0.07
FDNet	**97.62±0.24**	**97.66±0.21**	**97.63±0.24**	**99.62±0.04**	**97.61±0.23**

TABLE VII
CLASSIFICATION PERFORMANCES OF DIFFERENT MODELS ON AID DATASET

Model	Accuracy(%)	Precision(%)	Recall(%)	Specificity(%)	F1-score(%)
Resnet50 [9]	94.70±0.45	94.68±0.18	94.41±0.45	99.80±0.01	94.49±0.42
ConvNext [38]	92.80±0.15	92.81±0.23	92.40±0.20	99.74±0.01	92.50±0.22
GCSANet [39]	94.97±0.13	94.96±0.08	94.65±0.20	99.81±0.01	94.71±0.15
DBGA-Net [40]	94.80±0.45	94.78±0.53	94.58±0.39	99.81±0.02	94.61±0.45
ViT [16]	80.20±0.90	80.73±1.36	80.00±0.84	99.31±0.03	79.94±1.11
Swin-T [41]	89.57±0.08	89.50±0.04	89.31±0.01	99.62±0.01	89.24±0.02
MPViT [42]	92.83±0.37	92.94±0.37	92.56±0.36	99.74±0.01	92.60±0.37
EMTCAL [43]	93.25±1.20	93.23±1.17	92.93±1.13	99.75±0.04	92.97±1.20
MFC2Net [44]	93.68±0.37	93.63±0.35	93.33±0.36	99.76±0.01	93.40±0.33
SwiftFormer [45]	93.47±0.23	93.52±0.27	93.12±0.24	99.76±0.01	93.19±0.26
SeaFormer [46]	94.32±0.38	94.29±0.43	94.08±0.32	99.79±0.02	94.11±0.36
SMT [47]	93.43±0.42	93.44±0.42	93.15±0.36	97.48±2.29	93.20±0.39
FAT [30]	94.65±0.20	94.66±0.21	94.41±0.16	99.80±0.01	94.41±0.16
RMT [48]	94.67±0.23	94.71±0.29	94.37±0.26	99.80±0.01	94.47±0.28
WCNNs [22]	92.88±0.22	92.82±0.28	92.61±0.32	99.74±0.01	92.65±0.31
Wave-ViT [26]	93.10±0.10	93.25±0.15	92.80±0.01	99.75±0.01	92.89±0.15
WDTCNN [24]	93.92±0.18	93.97±0.19	93.64±0.22	99.77±0.01	93.70±0.19
FDNet	**95.92±0.13**	**95.90±0.13**	**95.77±0.17**	**99.84±0.01**	**95.79±0.17**

the hybrid architecture of CNN and ViT, FDNet's various evaluation indicators are better than them, indicating that FDNet extracts local and global information in the frequency domain and the strategy of integrating the advantages of

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CNN and ViT structures is more effective. Although the WCNNs based on a hybrid structure of frequency domain methods and neural networks has the smallest computational and parameter requirements in the compared networks, FDNet has higher accuracy than WCNNs on three datasets. It shows that FDNet adopts a better strategy in combining frequency domain methods and neural networks.

D. Visual Analysis

To assess the FDNet model's performance more intuitively, we use the experimental data that has the best classification accuracy for visual analysis. On the three datasets, the actual classification result of each category is obtained by calculating the confusion matrix of FDNet. Grad-CAM is used to draw a heat map, which visualizes the model's highlighted areas of interest in the image. The confusion matrix of FDNet on SIRI-WHU dataset is shown in Fig. 11. The confusion matrix of FDNet on RSSCN7 dataset is shown in Fig. 12, and the confusion matrix on AID dataset is shown in Fig. 13. From the three confusion matrices, it can be seen that the vast majority of scenes are fully recognized, and some scenes are misclassified due to small interclass differences in appearance compared with other scenes, such as between "grassland" images and "field" images and between "industrial area" images and "dense residential area" images. In addition, on SIRI-WHU dataset, "pond" as a local component of some scene classes can easily be confused with "river" or "park" scene. On RSSCN7 dataset, "river and lake" as a local component of some scene classes can easily be confused with "parking lot," "forest," or "industrial area" scene. On AID dataset, "playground" as a local component of some scene classes can easily be confused with "school" or "amusement park" scene.

On SIRI-WHU dataset, the heat maps of FDNet and some models are shown in Fig. 14. On RSSCN7 dataset, the heat maps of FDNet and some models are shown in Fig. 15. On AID dataset, the heat maps of FDNet and some models are shown in Fig. 16. As shown in the heat map legend in Fig. 14, when the heat map gradients from blue to green to yellow until it finally focuses on red, it reflects the model's attention distribution when processing images. As the color transitions from blue to red, attention gradually focuses on more prominent or important areas. As can be seen from Figs. 14–16, FDNet can capture more pixels that are beneficial to target object recognition. For industrial areas and parking lot scenes with dense target objects, FDNet can better capture the main characteristics of each scene. For rivers, lakes, playgrounds, and storage tank scenes with distinct image content, the edges and contour information extracted by FDNet are more obvious and pay close attention to the target object. Overall, FDNet recognizes scene images more accurately.

V. Conclusion

To effectively combine frequency domain methods with neural networks and achieve complementary advantages of different data processing methods, this article introduces wavelet transform to improve downsampling operations and proposes an effective RSSC solution, which focuses on feature extraction in the frequency domain. Based on this solution, the model FDNet is proposed. At the same time, according to the difference in information stored in high-frequency and low-frequency images, this article makes full use of the respective advantages of the CNN structure and ViT structure and proposes the MSFC block and MDFSA block. The MSFC block and MDFSA block are used for MSL and multidimensional global feature extraction in frequency domain images, respectively. These blocks help capture image information more comprehensively in the frequency domain. To verify the effectiveness of FDNet, we chose to conduct comparisons and ablation experiments on the SIRI-WHU dataset, RSSCN7 dataset, and AID dataset. Experimental results show that FDNet's classification performance is significantly better than other advanced methods participating in the comparison. On the one hand, FDNet provides a great example of using frequency domain methods as the backbone of neural networks. On the other hand, FDNet provides a new research direction for RSSC, which is to extract features only in the frequency domain. We believe that with further research on frequency domain methods, there may be further improvements in the effectiveness of feature extraction only in the frequency domain. In further work, we will continue to investigate image processing techniques in the frequency domain in-depth and enhance the model's generalization capacity. We will also consider extending the method to other downstream tasks, such as image segmentation to explore its potential application fields.

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Wei Wang received the B.S., M.S., and Ph.D. degrees in information and communication engineering from the National University of Defense Technology, Changsha, China, in 1997, 2003, and 2010, respectively.

He is currently a Professor with the School of Computer and Communication Engineering, Changsha University of Science and Technology, Changsha. His research interests include remote sensing image processing, computer vision, and deep learning.



Tianle Yang received the B.E. degree in Internet of Things engineering from Xuchang University, Xuchang, China, in 2022. He is currently pursuing the M.E. degree in electronic information with the School of Computer and Communication Engineering, Changsha University of Science and Technology, Changsha, China.

His research interests include remote sensing image processing and computer vision.



Xin Wang received the B.S. and M.S. degrees in information and communication engineering from Wuhan University of Technology, Wuhan, China, in 1998 and 2006, respectively.

She is currently a Lecturer with the School of Computer and Communication, Changsha University of Science and Technology, Changsha, China. Her research interests include signal processing, computer vision, and pattern recognition.

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